



The ideal neighbourhoods of successful ageing: A machine learning approach

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ABSTRACT

Ageing outcomes are shaped by not only health conditions, but also their interactions with the external environment. While the effects of some specific neighbourhood characteristics such as rurality on ageing have been evaluated in various studies, we still know little about the relative importance of particular natural and urban environments and how the impact varies at different stages of the ageing process. This article addresses these knowledge gaps by analysing survey data from 33 European countries using a machine learning method called multivariate regression trees (MRT). Multiple wellbeing indicators are combined to form an ageing profile for each individual in the survey. After studying these profiles using MRT, we find that generally the affordability of health facilities is a major determinant of life satisfaction, self-rated health condition and mental wellbeing for individuals in most age groups. Other important but age-specific determinants are neighbourhood safety and accessibility to cultural facilities and to green areas. In contrast, characteristics such as urbanity, transportation and air quality do not significantly influence ageing outcomes. Our findings lend support to the resources theory in explaining ageing outcomes and suggest that more resources may have to be directed to improve the affordability and quality of health care services, the policing services and the accessibility to cultural and green areas in order to achieve more favourable ageing outcomes.

1. Introduction

Like other assets, health is distributed unevenly in space. In the city of Rotterdam, life expectancy in Pernis, a borough on the west side of Waalhaven harbour, is 84 years. Cross the harbour, in Charlois, life expectancy is 6 years less (Municipal Health Service of Rotterdam-Rijnmond, 2016). Many studies have argued and found that health is associated with area of residence (Bernard et al., 2007; Curtis and Rees Jones, 1998; Ishikawa et al., 2015; Pérez et al., 2020; Macintyre et al., 2002; Marmot and Bell 2012; Wilson et al., 2020; Woods 2003). As older adults tend to spend more time in their neighbourhoods (Glass and Balfour 2003), neighbourhood characteristics might impact on ageing outcomes (Strobl et al., 2016; Sugiyama et al., 2009). This begs the questions: are there geographical determinants in shaping

successful ageing? If so, what are they?

Two major theories have been proposed to explain how neighbourhood environment might affect ageing outcomes: resources theory and theory of risk exposure.¹ The resources theory proposes that poorer ageing outcomes can be explained by neighbourhood deprivation (Lang et al., 2008). A wealth of empirical literature has shown that neighbourhood deprivation is a main driver of differential ageing outcomes in terms of life satisfaction, health status and mental wellbeing (Winterton et al., 2016; Yen et al., 2009). Resources can affect ageing outcomes in three different ways. First, older adults living in a disadvantaged neighbourhood are usually endowed with fewer resources to manoeuvre in order to age well (Lang et al., 2008). Studies have shown that older people's health and mental wellbeing is related to their access to health care services (Ryser and Halseth 2012), green and blue areas (Dempsey

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¹ The labels of resources theory and theory of risk exposure are rarely used and referred to in the literature. We organise the literature based on the main arguments in the studies.

et al., 2018; Finlay et al., 2015; Hartig et al., 2014), food stores and restaurants (Garipey et al., 2015), transportation services (He et al., 2020), recreational facilities (Liu et al., 2021), internet infrastructure (Khalaila and Vitman-Schorr 2018), among others. Second, ageing outcomes are constantly shaped by the living environment through its interactions with people's body conditions, state of mind and activities (Beard and Petitot 2010; Clarke and Nieuwenhuijsen, 2009). For instance, while the presence of urban green areas can have a positive effect on health and mental wellbeing due to the benefits of cleaner air and scenic view (Islam et al., 2012; Mansor et al., 2012), it may affect health indirectly by inducing more physical and social activities (Gheasi et al., 2019; Groenewegen et al., 2006). A similar relationship between public amenities, activities and ageing outcomes has been found in connection with, for example, pedestrian infrastructure, shops and leisure facilities, and public transportation (Day 2008; Michael et al., 2006). Third, a better neighbourhood helps older individuals to overcome living problems related to declined physical conditions and facilitates actions that lead to better living experiences (Loh et al., 2019). For example, a systematic literature review by Morley et al. (2017) reported that the availability of and proximity to integrated care services are associated with the prevention of functional decline. Studies by Cherrie et al. (2018), Clarke et al. (2014), Finlay et al. (2015) and Hartig et al. (2014) showed that access to the natural environment could have cumulative and long-term impact on older people's health and cognition.

Concurrently, many studies showed that various neighbourhood risk factors are correlated with poorer ageing outcomes (Lafortune et al., 2016). The association might be explained by three mechanisms. First, hazards and stressors in the neighbourhood environment might cause injuries and other negative health outcomes. Examples of these hazardous factors include pollution, crime and poorly designed housing and built environment. Regarding pollution, noise from traffic and neighbourhood is related to hypertension and depressive mood (Barceló et al., 2016; Niemann et al., 2006). Traffic-related air pollution is found to be connected to deteriorated cognitive functions, depressive and anxiety symptoms, and health problems such as asthma and stroke (Andersen et al., 2012; Pun et al., 2017; Tonne et al., 2014). Senior citizens who live in neighbourhoods with higher crime rates may display greater depressive symptom severity and other stress disorders (Baranyi et al., 2021; Benjet et al., 2019). Gilster (2014) reported that neighbourhood disorders and incivilities such as vandalism and deserted homes are associated with depression and anxiety. The second mechanism is related to inactivity. To minimise risk exposure, older people might avoid going out, thus reducing their physical and social activities, which may negatively affect their physical health and mental wellbeing (Choi and DiNitto 2016; Clarke and Gallagher 2013). Using the "Street View" feature of Google Earth, Clarke and Gallagher (2013) showed that accessible features of a place is related to outdoor mobility. Auais et al. (2017) found that fear of falling has a significant relationship with mobility. Since mobility limitations and reduced levels of leisure activities are correlated with depression and perceived health stress (Lee et al., 2012), hazards in neighbourhoods may contribute to worse health and mental wellbeing (Lyu et al., 2021). The third mechanism is related to functional decline in the ageing process. Neighbourhood risk factors may inhibit or reduce the speed of recovery from physical functional impairments (Latham and Williams 2015). For instance, an uneven ground may discourage older adults' physical activities, which can be important for continued improvement in functional health after a fall (Latham-Mintus 2020; Tiedemann et al., 2011).

While both resources and risk factors are likely to be relevant, the mere presence or absence of a single factor is insufficient to explain differential ageing outcomes. As pointed out by Hartig et al. (2014), the mechanisms by which environment might affect health outcomes are synergistic and multiple. One empirical challenge is to identify these combinations of factors and to examine how different resource and risk factors interact with individual conditions and activities. This study is designed to tackle this challenge.

This study aims to identify and characterise different ageing pathways and the associated ageing profiles using a novel machine learning method called multivariate regression trees.² This article contributes to the literature in several ways. First, in general, we do not know how many pathways exist and how they look like. Results from our analysis will provide an answer to these two questions. Second, while results from regression analyses may confirm that, for example, access to green area is positively related to mental wellbeing, we know relatively little about whether this factor is essential for most or only some of the pathways. Research questions that involve higher order interactions are not easily answered by a usual regression analysis (James et al., 2013). Using a novel machine learning method called multivariate regression trees (De'ath 2002), our study will be able to tell how prevalent a factor is in explaining certain age outcomes. Third, ways to successful ageing is likely to be multiple. While functional decay is inevitable in the ageing process, the presence or absence of one factor could be compensated by other factors, for example, through the acquisition of new life skills (Baltes and Carstensen 1996). At the policy level, our findings can inform policymakers what kind of community facilities might be needed to improve the welfare of older people with certain characteristics and allow compensations to take place.

The remaining part of our study is organised as follows. The next section provides more information on the machine learning approach adopted in our study as well as on the data used. Then the empirical research method is described, followed by a presentation of the modelling results. Next, an interpretation and discussion is offered, followed by some retrospective and prospective remarks.

2. Data and methods

The data in our analysis are derived from the Fourth European Quality of Life Surveys (EQLS) in 2016. The survey was administered by the European Foundation for the Improvement of Living and Working Conditions. It covers the 27 EU member states, as well as Albania, Montenegro, North Macedonia, Serbia, Turkey and the United Kingdom. Interviews were conducted between September 2016 and March 2017. About 37,000 adults, with at least 1,000 people per country, were surveyed. A stratified clustered sample design and random probability sampling procedures were used at all stages to ensure representativeness (Ahrendt et al., 2017). Given our research interests, we include only respondents who were aged 40 or above in this study. The subsample is further divided into five age groups: 40–49, 50–59, 60–69, 70–79, and 80 or above. This design allows us to address heterogeneity in the process of ageing. In total, 25,742 observations are used in the analysis.

Successful ageing is a multidimensional concept. Given our interests in health and general wellbeing, we use life satisfaction, self-rated health status and mental wellbeing as the measures of ageing outcomes. We measure life satisfaction and health status with self-rated indicators and measure mental wellbeing with the World Health Organisation's WHO-5 Index (World Health Organization, 1998). For brevity, we report results from a model with only three outcomes.³ For ease of comparison and interpretation, each indicator is rescaled and standardised to have a zero mean. The scale of some variables is reversed, too. Therefore, a positive value always means an above average performance.

² In this study, we call the combinations (or configurations) of individual characteristics, activities, resources and risk factors as ageing pathways and the combinations of ageing outcomes as ageing profiles.

³ Results from an expanded model with 14 additional outcomes are comparable.

The analysis includes 24 potential determinants of successful ageing.⁴ They are grouped into four categories: neighbourhood characteristics, activities and social support, body condition, and socioeconomic status. The descriptions of the variables are summarised in Table 1.

Neighbourhood characteristics are captured by twelve items in three categories: urbanity, facilities and environment. *Urban* is a categorical variable, which describes the urban environment of a location, according to the subjective perception of the respondents. The categories are (1) open countryside, (2) village or small town, (3) medium or large town, and (4) city or city suburb. The facilities category includes five variables: *culture*, *shops*, *transport*, *afhealth* (affordability of health care services), and *qhealth* (quality of health care services). They measure the accessibility of recreational and cultural facilities, shops, public transport, as well as the affordability and the quality of health care services. The environment category contains six variables: *green*, *air*, *noise*, *clean*, *traffic*, and *safe*. They measure whether the respondents had easy access to green areas and live in a quality environment.

Social activities are associated with physical health and mental health (Giordano et al., 2012; Nummela et al., 2008). The category of social activities and support has nine variables. They capture the social life and daily activities of the respondents. They measure how frequent the respondents attended religious services, participated in social activities or community services and had frequent contacts with friends and/or relatives. They also indicate whether they were part of a support network and could get help from friends and neighbours when needed. The body condition category has one variable. It indicates whether the respondent had chronic physical or mental health problems, illnesses or disabilities. Finally, the socioeconomic status category has four items: *gender*, level of *education*, *income* (quartile) and *pension*. The last variable captures the perceived quality of a national pension system.

To be sure, the above classification is not entirely unambiguous, as an indicator can have direct and indirect effects on multiple dimensions of ageing. For instance, the absence of chronic illnesses not only contributes to better perceived health level, but also the mental wellbeing of an individual and life satisfaction. To capture the multiplicity of ageing and the correlations between different factors, we use a multivariate regression tree analysis, introduced below, to model successful ageing outcomes.

3. Empirical model

As Hartig et al. (2014) points out, pathways to successful ageing are multiple and synergistic. The effect of one beneficial neighbourhood characteristic might be dependent on certain individual factors and neighbourhood characteristics. For instance, the health benefits of nature might be dependent on the physical fitness of a person and the quality of infrastructure. Older adults who have difficulty walking are likely to be benefited less from green spaces disconnected to their dwellings (Keskinen et al., 2018). Therefore, it is essential to know how different factors interact to produce an effect and what neighbourhood characteristics are more (or less) relevant to which population subgroups. One major empirical challenge is how to identify the constellation of factors – which we call pathways – that jointly determine successful ageing outcomes. While regression analysis is particularly good in estimating the marginal effect of an independent variable, it is less capable of detecting interaction effects that involve multiple variables (Athey and Imbens 2019). For example, a saturated model with three factors contains seven independent variables (i.e., A, B, C, AB, AC,

Table 1

Indicators and determinants of successful ageing.

Variable	Survey Question/Description
Life satisfaction	How satisfied would you say you are with your life these days? 0 (very dissatisfied) to 1 (very satisfied)
Health	In general, how is your health? 0 (very bad) to 1 (very good)
Mental wellbeing	WHO5-Mental Wellbeing Index. 0 (very bad) to 1 (very good)
Gender	Gender of the respondent
Edu	3 categories: Lower secondary or below; Upper secondary or post-secondary; Tertiary
Income	Percentile group of income in the county of residence (4 quartiles)
Pension	In general, how would you rate the quality of the state pension system in your country? (Regional average; 1 (very poor) – 10 (very high))
Spiritual	How frequently do you attend religious services, apart from weddings, funerals or christenings? 5 categories: everyday/almost every day; at least once a week; one to three times a month; less often; never
Socact	How frequently do you participate in social activities of a club, society, or an association? 5 categories: everyday/almost every day; at least once a week; one to three times a month; less often; never
Contact	Do you have contact with friends/family everyday? (Yes/No)
Live alone	Do you live alone? (Yes/No)
Has child	Do you have child in household or elsewhere?
Chronic	Do you have any chronic (long-standing) physical or mental health problem, illness or disability? By chronic (long-standing) I mean illnesses or health problems which have lasted, or are expected to last, for 6 months or more. (Yes/No)
Volunteer	How often did you do unpaid voluntary work for community and social services? 4 categories: every week; every month; less often/occasionally; not at all
Support	Can get help from friend, neighbour, or someone else when ill (Yes/No)
Urban	Would you consider the area in which you live to be ... 4 categories (open countryside/village or small town/medium to large town/city or city suburb)
Culture	Thinking of physical access, distance, opening hours and the like, how easy or difficult is your access to cinema, theatre or cultural centre? 4 categories: very difficult; rather difficult; rather easy; very easy
Shops	Thinking of physical access, distance, opening hours and the like, how easy or difficult is your access to grocery shop or supermarket? 4 categories: very difficult; rather difficult; rather easy; very easy
Transport	Thinking of physical access, distance, opening hours and the like, how easy or difficult is your access to public transport? 4 categories: very difficult; rather difficult; rather easy; very easy
Green	Thinking of physical access, distance, opening hours and the like, how easy or difficult is your access to recreational or green areas? 4 categories: very difficult; rather difficult; rather easy; very easy
Noise	Please think about the area where you live now – I mean the immediate neighbourhood of your home. Do you have major, moderate or no problems with noise?
Air	Please think about the area where you live now – I mean the immediate neighbourhood of your home. Do you have major, moderate or no problems with air quality?
Clean	Please think about the area where you live now – I mean the immediate neighbourhood of your home. Do you have major, moderate or no problems with litter or rubbish on the street?
Traffic	Please think about the area where you live now – I mean the immediate neighbourhood of your home. Do you have major, moderate or no problems with heavy traffic in your immediate neighbourhood?
Afhealth (affordability of health care services)	How easy or difficult would it be for you to cover expenses for GP, family doctor or health centre

(continued on next page)

⁴ In an initial endeavour, we have expanded the analysis to include as many as 34 successful ageing indicators and eight ageing outcomes to cover other dimensions of successful ageing, such as psychological resilience, social integration, self-esteem, financial security, among others. The findings are largely similar.

Table 1 (continued)

Variable	Survey Question/Description
	services? 5 categories: very easy, rather easy, neither easy nor difficult; rather difficult; very difficult
Qhealth (quality of health care services)	In general, how would you rate the quality of health services in your country? 1 (very poor) - 10 (very high)
Safe	To what extent do you agree or disagree with the following statements? (5-point Likert scale). Average of the responses to the two statements: I feel safe when I walk alone after dark. I feel safe from crime when I am at home alone at night.

BC, ABC). The size of a model explodes when the number of factors increases. (In our case, we have 24 variables. Many of them are categorical variables.) The problem of multi-collinearity also makes the task of dependency identification prohibitive.⁵ The method of regression tree is capable of identifying interaction effects among multiple variables and fits for our research purpose.⁶ More importantly, results from a regression tree can tell us how various pathways (i.e., sets of independent variables) look like, what neighbourhood characteristics are important in a particular pathway, and which pathway is associated with better (or worse) ageing outcomes for individuals in a certain age group. Multivariate regression trees (MRT), a technique originating and used mainly in ecology, is an extension of the regression tree method (De'ath 2002). It models multiple dependent variables, capturing the multidimensional nature of ageing (Bowling and Dieppe 2005; Depp and Jeste 2006; Ryff 1989). In our analysis, we combine three ageing indicators to form an ageing profile. These profiles, one for each individual, are then classified by various independent variables using the MRT method.

We create an ageing profile for each individual based on the self-rated performance in the three successful ageing indicators listed in Table 1. Fig. 1 below shows some examples of a profile based on the national average of the indicators (after standardisation). People may have a similar profile due to the similarities of their living environments and demographic characteristics. Admittedly, heterogeneity exists and an older adult who is physically ill can be psychologically strong. But because of the correlations between indicators, some profiles are likely to be more common than others. For example, a person who has chronic illnesses or disabilities is more likely to be financially stressed. One objective of this study is to characterise the archetypes and then identify factors that predict these profiles. Major challenges are that in general we do not know how many archetypes exist and how they look like. To tackle these challenges, we apply a regression tree method for unsupervised (machine) learning (Breiman et al., 1984) to characterise typical profiles. The method iteratively partitions a sample into two sub-samples by minimising similarities within each subsample and maximising the dissimilarities between the two subsamples in each iteration. Thus, similar to a cluster analysis, the method identifies common characteristics shared by individuals in the same cluster.

A regression tree analysis has two steps: partition and pruning (James et al., 2013). Initially, the whole sample is randomly divided into two subsamples: a training set and a test set. To grow a tree, one starts with the training set. For each partition, observations in the training set are divided into two non-overlapping parts (called nodes) by an independent variable. Let's take the variable *income*, which has four possible

values, as an example. One can partition the sample into two groups, one with an income below the national median and one above. The choices of the partition variable and the threshold are non-random but guided by a rule: in each partition, we will choose the variable and the threshold such that the within-group sums of squared errors in the nodes are minimised. That is,

$$\min \sum_{i \in R_1} (y_i - y_{R_1})^2 + \sum_{i \in R_2} (y_i - y_{R_2})^2 \quad (1)$$

where i is the index of observations, y_i the observed value of i , y_{R_j} the mean of y for all the observations in group j , and R_j a set consisting of observations in node j . The same procedure is repeated for each resulting node until a stopping criterion is reached. Partition will stop when the number of observations in a node becomes small (e.g., three). This criterion is usually applied when the sample size is small. For the other stopping criterion, partition stops when the reduction of the errors by further growing the tree is small. This happens when no independent variable can sufficiently outperform the others in improving predictions. There is a complete tree when a stopping criterion is reached and the tree stops growing.

The complete tree will overfit the data in the training set, but will give a poor fit to the data in the test set. To avoid this overfitting problem, we move to the second step: pruning. To prune a tree, the regression tree technique uses k -fold cross-validation (Borcard et al., 2011). The idea behind cross-validation is that the data are divided into k folds. In each run, we retain $1/k$ of the observations as the test examples and use the remaining observations to build a tree. We use the data in a test set to assess the performance of the tree based on the cross-validated relative error (CVRE) of the tree, which is equal to:

$$CVRE = \frac{\sum_{i=1}^n \sum_{j=1}^p (y_{ij(k)} - \widehat{y}_{j(k)})^2}{\sum_{i=1}^n \sum_{j=1}^p (y_{ij} - \bar{y}_j)^2} \quad (2)$$

where $y_{ij(k)}$ is an observation in test set k and $\widehat{y}_{j(k)}$ the predicted value for the observation. The denominator is the total sum of squares. As such, the CVRE can be interpreted as the fraction of variance unexplained by the tree. The CVRE is calculated at every partition for each tree, and the process is repeated k times. The standard errors (SE) of the CVRE values are calculated. The final tree is the one which has less nodes and has the CVRE just below the minimal CVRE plus one standard error of the CVRE values (i.e., the one standard error rule in Breiman et al. (1984)). In this study, since we want to identify relevant profiles and the associated factors that lead to these profiles, sometimes we selected a slightly more complicated model with a smaller CVRE.

So far we assumed that y in equation (1) is a scalar. In our analysis, since we have more than one indicator of successful ageing, we extend the univariate setting to a multivariate one, following the multivariate regression trees (MRT) method by De'ath (2002). The procedure and principles of growing and pruning a tree are similar to what was described. The only difference is that in a multivariate setting, y_i is a vector and y_{R_j} the centroid of the vectors in R_j , and the sum of squared differences in (1) becomes the sum of squared Euclidean distances between vector y_i and centroid y_{R_j} . The multivariate regression tree analysis has mainly been used in ecology (e.g., Brodie et al. (2007); Jafino and Kwakkel (2021)). This study introduces the method to gerontology to characterise a neighbourhood where people with a certain wellbeing profile are populated.

4. Results

4.1. Descriptive results

Fig. 1 displays the country averages of the wellbeing indicators. For ease of comparison, indicators are sorted in descending order using Austria as a benchmark. Fig. 1 reveals that people living in western and

⁵ While collinearity may cause problems in regression analysis, it is not a major concern in the regression tree method (Loh 2006). We also calculated the Pearson's and Spearman's correlation coefficient of all 27 variables. The absolute values of the coefficients never exceed 0.6. Due to the size of the matrices, they are not reported.

⁶ See Buchner et al. (2017), Lemon et al. (2003) and Wang et al. (2019) for applications in the public health literature.

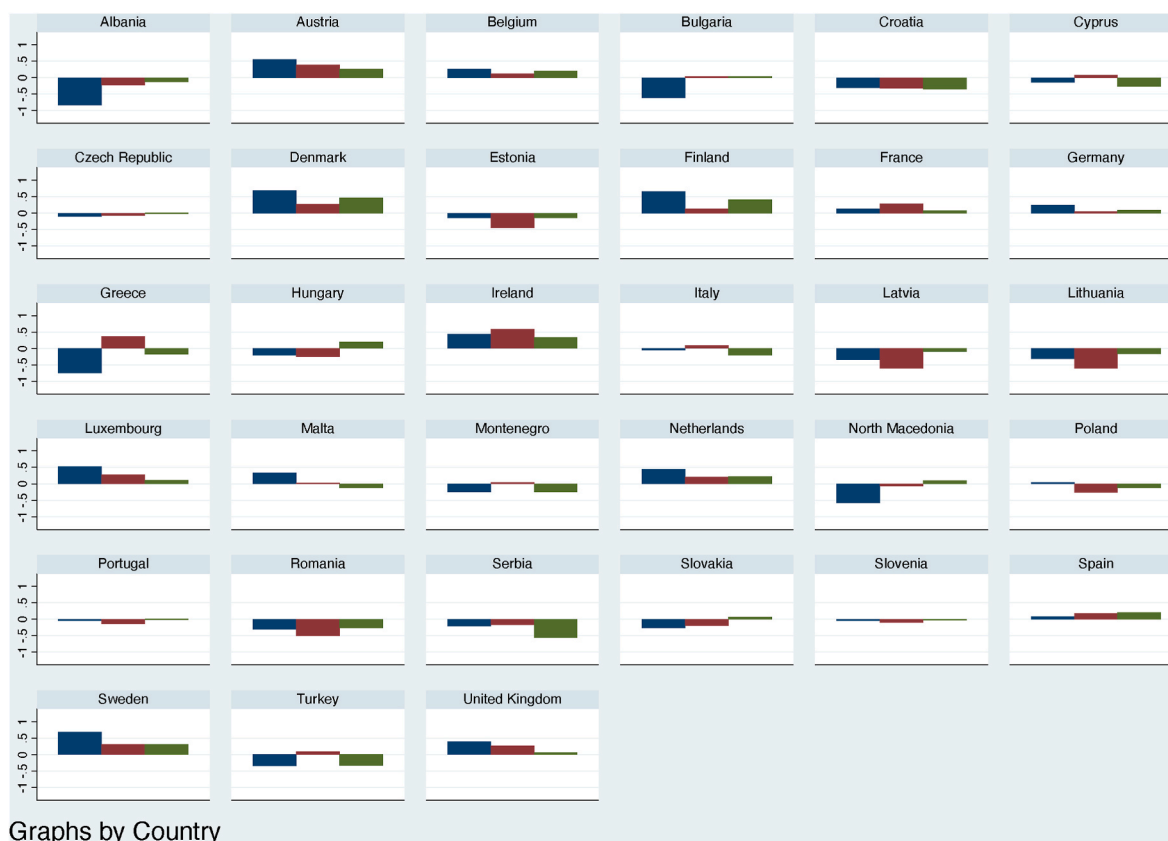


Figure 1. Successful ageing profiles (country average), by surveyed country. Notes: From left to right: life satisfaction, physical health, mental wellbeing. Standardised mean values reported (on the y-axis).

northern Europe have a higher self-perceived wellbeing in almost all three measured outcomes. In contrast, middle aged and older adults living in Albania, Croatia and Romania have lower (self-rated) wellbeing outcomes. Interestingly, while adults in Greece and North Macedonia have a lower level of life satisfaction than their counterparts in Italy or Portugal, they score above average in health (Greece) and mental wellbeing (North Macedonia). Geographical clusters seem to exist as well. The Nordic countries (Denmark, Finland and Sweden), the Baltic states (Latvia and Lithuania), and Central Europe countries (Hungary and Slovakia) share some similarities in terms of their profiles, although some distinct differences can still be spotted within clusters (e.g., Bulgaria and Romania).

Profiles not only differ across countries but also between age cohorts. We disaggregate the profiles into five age groups: 40–49, 50–59, 60–69, 70–79, 80 or above. Some patterns are identified from Figs. 2–4. First, except perhaps Denmark, older people aged 70 or above have reported a lower self-rated health status than people aged 50–59 and 60–69 (Fig. 2). Although the same trend is observed among older people living in Ireland, the Netherlands and the UK, people above 70 in these three countries maintain an above-average rating. Second, regarding mental wellbeing (Fig. 3), a clear decreasing trend can be identified in some countries (e.g., Croatia, Italy, Lithuania and Poland). Finally, for life satisfaction (Fig. 4), there are substantial variations across countries even within a regional block. For example, the indicator is relatively stable across age groups in Cyprus, Czechia and Latvia. In contrast, older people tend to have a lower level of life satisfaction in Bulgaria and Portugal. Finally, scoring low in one dimension does not necessarily affect overall life satisfaction. For example, while older people in Malta have scores below the sample averages in self-rated health status and mental wellbeing, the life satisfaction scores are well above the averages.

4.2. Results from multivariate regression trees analysis

Table 2 lists the factors that predict the relative performance of the three successful ageing indicators by age cohort. The regression tree for the 40–49 cohort contains seven end nodes. The numbers under the column “Life” are the average values of the life satisfaction indicator for individuals with a pathway specified in each row. The “pathway” column records the tree path that leads to an end node. The row averages of the three indicators are reported under the column “average”. The numbers can be interpreted as the average level of general wellbeing for people with the listed profile. For example, individuals who have a profile more similar to the one specified in row 1 than to the profile in row 2 tend to have chronic illnesses and tend to find the health care services (HCS) in their neighbourhoods very or rather difficult to afford. There are 299 individuals aged between 40 and 49 who have a profile similar to profile 1. The average life satisfaction score for this group is -0.64 , which is the lowest among the seven clusters in that age group. The graphical representation of Table 2 can be found in Figs. 5–9.

The regression tree for the 40–49 cohort contains seven end nodes (Fig. 5). For people aged 40–49, of the 24 variables evaluated, the existence of chronic illnesses or disabilities, perceived HCS affordability, HCS quality, perceived safety, the level of education and income (quartiles) are crucial classifiers of successful ageing profiles. As the first identified variable that splits the samples, the existence of chronic conditions is the best discriminator of profiles. People with chronic illnesses or disabilities are predicted to have a lower level of general wellbeing. Among those who have chronic illnesses, the second best discriminator is the affordability of HCS. When compared rows 1 and 2 in Table 2 (or the average height of the bars for P1 and P2), the average of the three ageing indicators is substantially lower among people who consider the HCS difficult to afford. Thus the affordability of the services not only has an impact on physical health outcomes but also mental

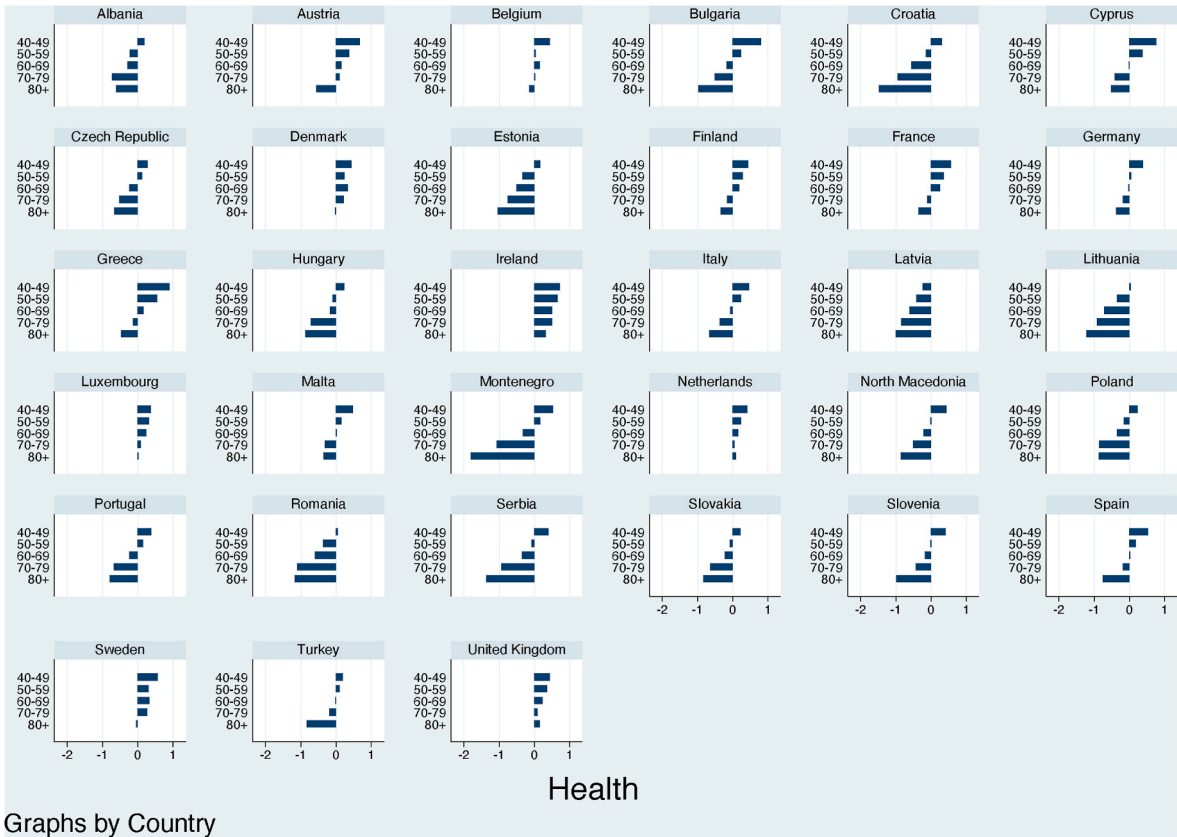


Fig. 2. Self-rated health score across countries and age groups. Notes: Standardised mean values reported.

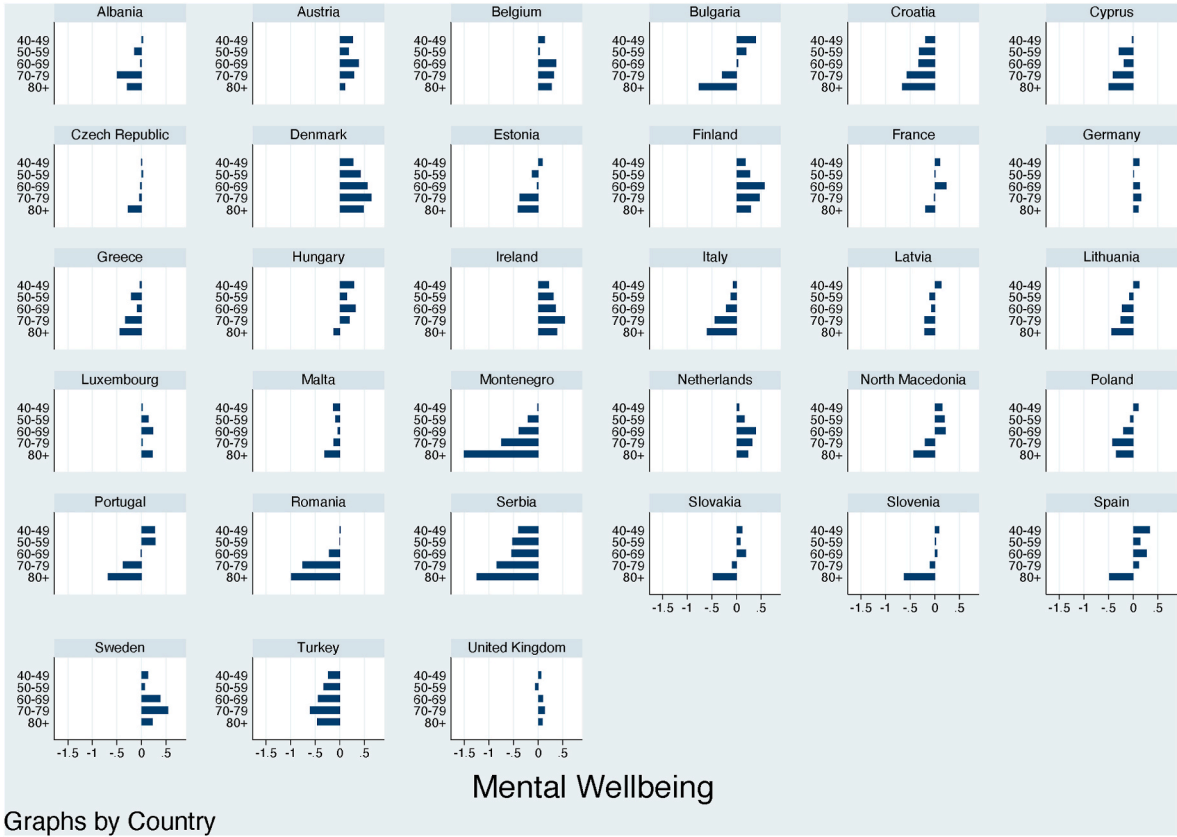


Fig. 3. Mental wellbeing across countries and age groups. Notes: Standardised mean values reported.



Fig. 4. Life satisfaction across countries and age groups. Notes: Standardised mean values reported.

wellbeing and life satisfaction. Among those who do not have any chronic illnesses, the second best discriminator is the quality of HCS, followed by perceived safety of the area, levels of education and income.

In general, among all of the evaluated environmental factors, the affordability and the quality of HCS and perceived safety are the factors most relevant to wellbeing outcomes for people aged between 40 and 49. Relative to these two factors, factors such as air quality, accessibility to public transport, recreational facilities and green areas are not major discriminators of wellbeing for people in this age group. Second, affordability of HCS seems to have a greater power in determining general wellbeing than the quality of the services for people with poorer health. When compared P1 with P2 in Fig. 5, by making the services more affordable, the difference in the indicator values is noticeable. Finally, from the tree structure, we can conclude that income level plays a role mainly for people without chronic illnesses or disabilities. When comparing P5 with P6, it is noticed that people with a lower income have a lower level of wellbeing (average = 0.29 versus 0.51 in Table 2).

The profiles for people aged 50–59 are summarised in the second panel in Table 2 and Fig. 6. The results can be interpreted in a similar way. We highlight some important findings in the following. First, similar to people in their 40s, the existence of chronic illnesses is the most critical factor that determines wellbeing. Second, safety and education are no longer crucial in determining wellbeing. Instead, the quality of the state pension systems and social activities enter the scene. Social activities matter the most for people with chronic illnesses and a lower level of wellbeing (P1–P3). In contrast, the quality of state pension is more relevant to people with a relatively high level of general wellbeing and at places where health care services are affordable (P7–P9). Third, HCS affordability is a concern for people without chronic illnesses as well (P4–P9).

The profiles for people aged 60–69 are summarised in Fig. 7. Again, the variable *chronic* is the most important discriminator. HCS affordability and quality, pension and income remain key determinants of

ageing outcomes. Except HCS, neighbourhood characteristics such as noise, shopping and recreational facilities are not major discriminators of wellbeing for people in this age group either. Affordability becomes clearly more relevant and is a major discriminator related to most of the profiles (P3–P8). A quality pension system also becomes more important for people who have chronic health problems (P1–P5). Finally, it is interesting to note that the average scores for the general and mental wellbeing between profiles 5 and 7, though distinct in their profiles, are similar (0.13 and 0.15 for the average scores; 0.19 and 0.19 for mental wellbeing).

Regarding people aged 70–79 (Fig. 8), health conditions and HCS affordability are again on top of the list, and so does pension quality. However, the neighbourhood environment becomes more important for older adults. Among the evaluated characteristics, safety of neighbourhoods and the accessibility of cinema, theatre and other cultural facilities are the most important, especially for those with health complaints (P1–P2). In fact, those living in an unsafe neighbourhood and with poor access to the cultural facilities have the lowest wellbeing scores among the six identified profiles (P1–P3). If we compare profiles 1 and 2, we found that cultural facilities have an impact on all three dimensions. The major difference between P3 and P4 is pension quality. (Safety is not a major concern in P4 and P3 has the highest scores in all three dimensions.) While pension quality may be associated with financial and mental stress, its effect on physical health seems to be less substantial.

The tree for people aged 80 or above are the most complicated (Fig. 9), reflecting a high level of heterogeneity among people in this group. If we take the age group average (−0.28) as the reference point, our results suggest that people age less well in many different ways, as there are seven out of 11 profiles with an average below −0.28. Free from major health problems, again, is the most decisive factor, but alone it is insufficient; for example, the average is −0.73 for P8. Social activities, affordable HCS and access to green areas are other essential complementary factors for successful ageing. Financial and physical

Table 2

Pathways to various successful ageing profiles, by age group.

	Pathway	N	Avg.	Life	Health	Mental
Group 1 (Aged 40–49)						
1	Chronic (Yes), afhealth (<3)	299	−0.64	−0.60	−0.64	−0.67
2	Chronic (Yes), afhealth (≥3)	986	−0.16	0.04	−0.32	−0.21
3	Chronic (No), qhealth (<7)	2306	0.13	−0.16	0.52	0.02
4	Chronic (No), qhealth (≥7), safe (<0.6875)	566	0.18	0.01	0.47	0.05
5	Chronic (No), qhealth (≥7), safe (≥0.6875), edu (<Tert), income (<Q3)	488	0.29	0.08	0.61	0.18
6	Chronic (No), qhealth (≥7), safe (≥0.6875), edu (<Tert), income (≥Q3)	538	0.51	0.36	0.77	0.39
7	Chronic (No), qhealth (≥7), safe (≥0.6875), edu (=Tert)	765	0.65	0.60	0.92	0.42
Group 2 (Aged 50–59)						
1	Chronic (Yes), socact (=1), income (<Q3)	683	−0.80	−0.75	−0.99	−0.67
2	Chronic (Yes), socact (=1), income (≥Q3)	405	−0.38	−0.26	−0.55	−0.33
3	Chronic (Yes), socact (>1)	773	−0.11	0.10	−0.33	−0.12
4	Chronic (No), afhealth (<4), qhealth (<6), income (<Q3)	308	−0.33	−0.79	0.12	−0.33
5	Chronic (No), afhealth (<4), qhealth (<6), income (≥Q3)	277	0.02	−0.26	0.36	−0.05
6	Chronic (No), afhealth (<4), qhealth (≥6)	932	0.13	−0.02	0.32	0.08
7	Chronic (No), afhealth (≥4), pension (<4.747), income (<Q4)	490	0.06	−0.24	0.28	0.16
8	Chronic (No), afhealth (≥4), pension (<4.747), income (≥Q4)	321	0.36	0.16	0.49	0.42
9	Chronic (No), afhealth (≥4), pension (≥4.747)	1659	0.44	0.39	0.62	0.31
Group 3 (Aged 60–69)						
1	Chronic (Yes), pension (<5.308), income (<Q3)	807	−0.72	−0.65	−0.96	−0.55
2	Chronic (Yes), pension (<5.308), income (≥Q3)	530	−0.27	−0.10	−0.55	−0.16
3	Chronic (Yes), pension (≥5.308), afhealth (<3)	160	−0.50	−0.18	−0.90	−0.41
4	Chronic (Yes), pension (≥5.308), afhealth (≥3), income (=1)	175	−0.27	0.09	−0.59	−0.33
5	Chronic (Yes), pension (≥5.308), afhealth (≥3), income (>1)	559	0.13	0.45	−0.26	0.19
6	Chronic (No), afhealth (<5), qhealth (<5)	512	−0.19	−0.52	0.05	−0.11
7	Chronic (No), afhealth (<5), qhealth (≥5)	1925	0.15	0.08	0.20	0.19
8	Chronic (No), afhealth (=5)	1366	0.47	0.42	0.48	0.51
Group 4 (Aged 70–79)						
1	Chronic (Yes), pension (<5.517), safe (<0.5625), culture (=1)	115	−1.29	−1.08	−1.59	−1.19
2	Chronic (Yes), pension (<5.517), safe (<0.5625), culture (≥1)	292	−0.81	−0.58	−1.09	−0.76
3	Chronic (Yes), pension (<5.517), safe (≥0.5625)	901	−0.50	−0.26	−0.88	−0.37
4	Chronic (Yes), pension (≥5.517)	768	−0.14	0.28	−0.61	−0.08
5	Chronic (No), afhealth (<5)	1235	0.04	−0.01	0.04	0.08
6		759	0.52	0.53	0.44	0.60

Table 2 (continued)

	Pathway	N	Avg.	Life	Health	Mental
Chronic (No), afhealth (=5)						
Group 5 (Aged 80 or above)						
1	Chronic (Yes), socact (=1), pension (<4.499), green (<3)	107	−1.29	−0.93	−1.66	−1.27
2	Chronic (Yes), socact (=1), pension (<4.499), green (≥3)	229	−0.75	−0.41	−1.20	−0.63
3	Chronic (Yes), socact (=1), pension (≥4.499), afhealth (<4)	172	−0.73	−0.33	−1.13	−0.73
4	Chronic (Yes), socact (=1), pension (≥4.499), afhealth (≥4), safe (<0.5625)	54	−0.75	−0.21	−1.12	−0.92
5	Chronic (Yes), socact (=1), pension (≥4.499), afhealth (≥4), safe (≥0.5625)	174	−0.30	0.10	−0.78	−0.23
6	Chronic (Yes), socact (>1), afhealth (<4)	56	−0.44	−0.17	−0.93	−0.23
7	Chronic (Yes), socact (>1), afhealth (≥4)	179	0.12	0.51	−0.36	0.21
8	Chronic (No), socact (=1), afhealth (<4), green (<3)	47	−0.73	−0.66	−0.65	−0.87
9	Chronic (No), socact (=1), afhealth (<4), green (≥3)	140	−0.22	−0.17	−0.24	−0.25
10	Chronic (No), socact (=1), afhealth (≥4)	277	0.16	0.20	0.05	0.24
11	Chronic (No), socact (>1)	252	0.47	0.56	0.38	0.48

Notes: afhealth (affordability of health care services): 1 (very difficult) to 5 (very easy), qhealth (quality of health care services): 1 (very poor) - 10 (very good), safe: 1 (feel safe) - 5 (feel unsafe), socact (social activities): 1 (never) - 5 (every day or almost every day), pension: 1 (very poor quality) - 10 (very high quality), culture: 1 (very difficult) - 4 (very easy), green: 1 (very difficult) - 4 (very easy).

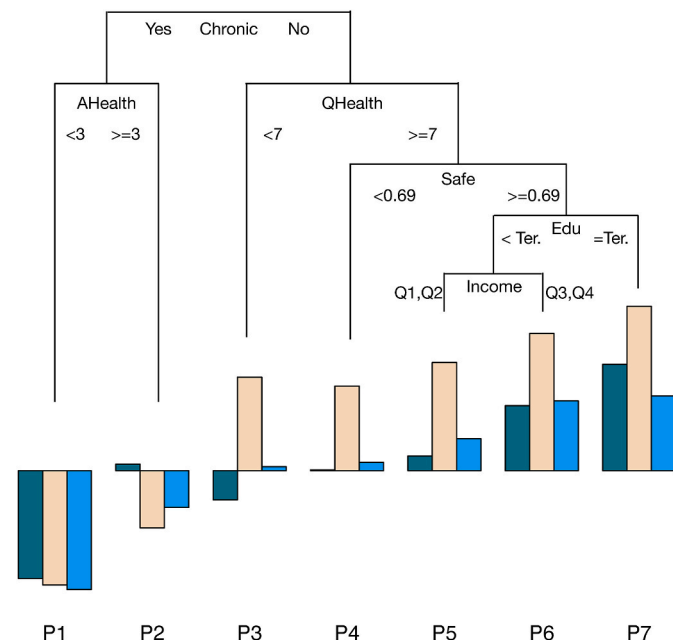


Fig. 5. Paths to various successful ageing profiles (40–49). Notes: From left to right: life satisfaction, physical health, mental wellbeing. Standardised mean values reported (on the y-axis). afhealth (affordability of health care services): 1 (very difficult) to 5 (very easy), qhealth (quality of health care services): 1 (very poor) - 10 (very good), safe: 1 (feel safe) - 5 (feel unsafe).

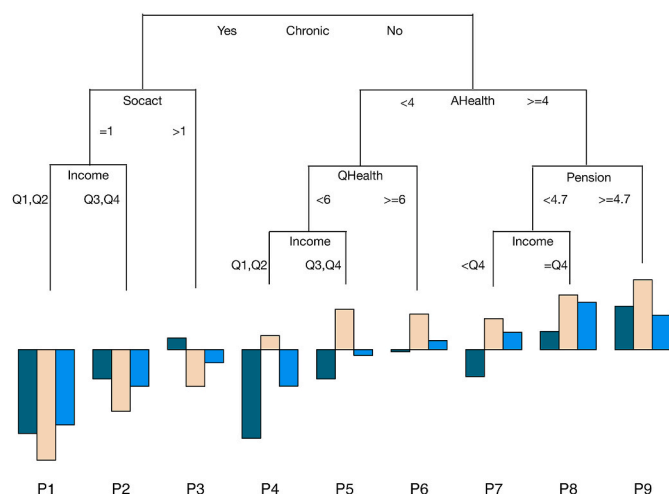


Fig. 6. Paths to various successful ageing profiles (50–59). Notes: From left to right: life satisfaction, physical health, mental wellbeing. Standardised mean values reported (on the y-axis). afhealth (affordability of health care services): 1 (very difficult) to 5 (very easy), qhealth (quality of health care services): 1 (very poor) - 10 (very good), socact (social activities): 1 (never) - 5 (every day or almost every day), pension: 1 (very poor quality) - 10 (very high quality).

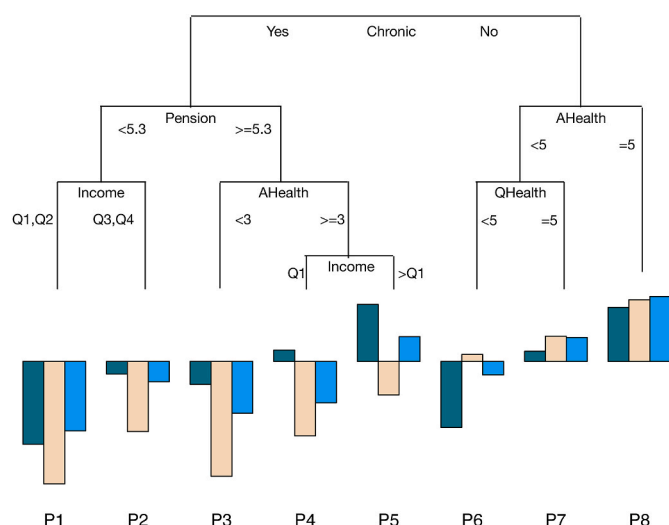


Fig. 7. Paths to various successful ageing profiles (60–69). Notes: From left to right: life satisfaction, physical health, mental wellbeing. Standardised mean values reported (on the y-axis). afhealth (affordability of health care services): 1 (very difficult) to 5 (very easy), qhealth (quality of health care services): 1 (very poor) - 10 (very good), pension: 1 (very poor quality) - 10 (very high quality).

security are related to general wellbeing, but their effects are quite marginal if we compare P2, P3 and P4. In contrast, the effect of HCS affordability is more pronounced for people engaging with social activities (P6 and P7). Furthermore, if we compare P1 and P2, high accessibility to green areas has shown to have a significant impact on all three outcome indicators.

A further comparison between predictors across age groups can also reveal the relevancy of a factor across age groups. Table 3 provides this information for comparison. The vitality of HCS affordability have been identified in all age groups. The quality of HCS is another decisive neighbourhood characteristic for people in multiple age cohorts. While shopping facilities, transportation system, location (e.g., urban or rural areas) and complaints related to noise and cleanliness are not particularly relevant in predicting ageing outcomes, physical security and accessibility to cultural facilities and green spaces are three of the

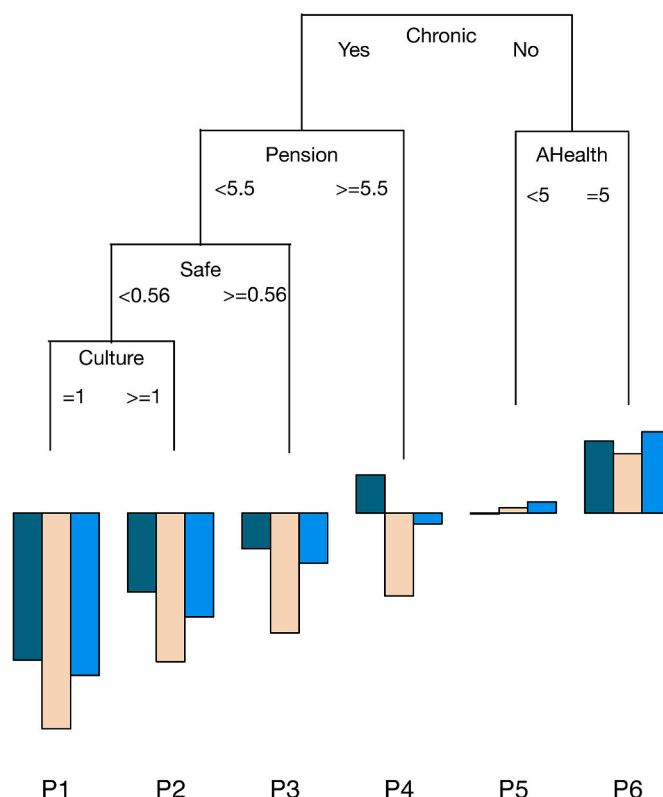


Fig. 8. Paths to various successful ageing profiles (70–79). Notes: From left to right: life satisfaction, physical health, mental wellbeing. Standardised mean values reported (on the y-axis). afhealth (affordability of health care services): 1 (very difficult) to 5 (very easy), safe: 1 (feel safe) - 5 (feel unsafe), pension: 1 (very poor quality) - 10 (very high quality).

neighbourhood characteristics most relevant for people aged 70 or above.

5. Discussion

Our results lead to a range of interesting reflections and interpretations. First, existing studies commonly contain only a small set of neighbourhood characteristics to test specific hypotheses. When different types of neighbourhood characteristics are taken into consideration, some of the commonly identified place-based predictors do not significantly predict successful ageing outcomes. According to our findings, location does not seem to systematically affect ageing outcomes. Urbanisation and some of the associated environmental factors such as accessibility to public transport, noise and air quality do not have substantial impact on successful ageing outcomes either. This is contrary to some recent studies; for example, Walsh et al. (2020) on rurality, Alidoust et al. (2019) on public space, Shi and Yu (2020) on air pollution, and He et al. (2020) on public transportation. Consistent with Finlay et al. (2015) and Liu et al. (2021), access to green areas and recreational facilities are important factors that contribute to better ageing outcomes, but we found that they are relevant mainly to people in certain age groups.

Second, regression analysis focuses on the marginal effect of individual factors. But as the results of our models show, the effect of one factor is usually dependent on the existence of other factors or conditions. For example, Ennsle and Kabisch (2020) propose that green space has beneficial effects on mental health and self-rated health. However, similar to Hooper et al. (2020), while we found supportive evidence for the positive role of nature on ageing, the effect is more evident to people in a later stage of a life cycle. Furthermore, our analysis also reveals that the effect is more pronounced for people with little social interactions

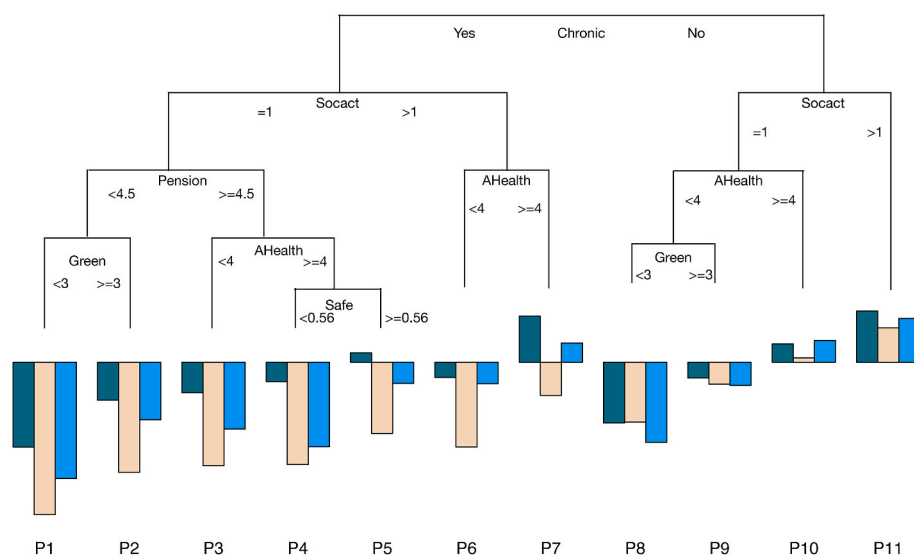


Fig. 9. Paths to various successful ageing profiles (80 or above). Notes: From left to right: life satisfaction, physical health, mental wellbeing. Standardised mean values reported (on the y-axis). affhealth (affordability of health care services): 1 (very difficult) to 5 (very easy), safe: 1 (feel safe) - 5 (feel unsafe), socact (social activities): 1 (never) - 5 (every day or almost every day), pension: 1 (very poor quality) - 10 (very high quality), culture: 1 (very difficult) - 4 (very easy), green: 1 (very difficult) - 4 (very easy).

Table 3
Discriminators of successful ageing profiles, by age group.

Age Cohort	Factors
40–49	Chronic, Affordability of Health Care Services, Quality of Health Care Services, Income, Safe, Education
50–59	Chronic, Affordability of Health Care Services, Quality of Health Care Services, Income, Pension, Social Activities
60–69	Chronic, Affordability of Health Care Services, Quality of Health Care Services, Income, Pension
70–79	Chronic, Affordability of Health Care Services, Pension, Safe, Culture
80 or above	Chronic, Affordability of Health Care Services, Pension, Safe, Green, Social Activities

and lower ageing outcomes. Therefore, as Figs. 5–9 demonstrate, the effect of nature (and other factors) does not work in isolation of other factors. The trees show various dependency structures clearly.

Health-related variables seem to be the most important determinants of ageing outcomes. Variables such as *chronic*, *afhealth* and *qhealth* are almost always on the list and they are relevant for people of various age groups (see Table 3). This is consistent with a finding reported by Bowling (2006). However, while health problems are the most influential factors in determining ageing outcomes, their effects on ageing can be, at least partially, moderated by quality and affordable health care services, as argued by Baltes and Carstensen (1996) and shown empirically by Or et al. (2005), Aakvik and Holmås (2006) and Zhang and Gao (2021).

Relatedly, adverse health conditions can be compensated by not only better neighbourhood facilities but also individual actions. In other words, agency also matters. As Fig. 9 shows, participation in social activities, for example, can have an impact on multiple indicators and “compensates” wellbeing loss due to poorer health. As another example of compensation, the relationship between urbanisation and ageing outcome is not monotonic, but contingent on a diverse set of factors (Beard and Petitot 2010). Older people living in rural areas could be more socially isolated, have poorer access to HCS, public transport and other supports for their daily lives (Walsh et al., 2020). On the other hand, they usually have better access to green areas, leading to more exercises and hence better mental and physical health outcomes (Andersson et al., 2019). The quality of social interactions in rural areas could be stronger and far more important than the quantity of them, conditional on some other neighbourhood characteristics (Van den Berg et al., 2017). All these factors complement and compensate for the losses in different aspects of life.

Fifth, as most of the identified factors are related to neighbourhood qualities and facilities, our study finds more support to the resources theory than the theory of risk exposure. Risk factors seem to be less relevant, as safety is the only risk factor showing up in our models. If older adults living in disadvantaged and under-resourced neighbourhoods tend to have poorer ageing outcomes (Cassarino and Setti 2015; Godhwani et al., 2019; Lang et al., 2008), then what kind of resources seem to be more important? According to our findings, inferior ageing outcomes can be related to the deprivation of health resources, low quality health care services, and inadequate access (maybe due to poor supporting infrastructure) to green areas, and cultural and recreational facilities. Our study also contributes to the resources theory by adding that, while some resources are available, their affordability can play a critical role in determining ageing outcomes. This issue is particularly relevant when inequality among older adults has become increasingly acute in Europe (Olivera 2020).

Finally, successful ageing is often seen as a product of individual level factors such as genetics (Lio et al., 2004), cognition (Castro-Lionard et al., 2011), psychology (Bowling and Iliffe 2011), physical health (Kalache and Kickbusch 1997) and financial security (Olivera and Tournier 2016). Though important, these views are limited as they tend to ignore various constraints imposed by the external living environment (Kuh et al., 2014) and underestimate the effects of the external environment on induced physical and social activities that shape ageing outcomes (Shirazi 2019). Nevertheless, as our analysis shows, the role of environment on shaping people’s behaviours remains a contested topic. A systematic review by Van Cauwenberg et al. (2011) reveals that, for example, the effects of urban characteristics (e.g., walkability) on human activities (e.g., walking) are more limited than expected. In our study, we found that the positive role of nature is also more pronounced among people with little social interactions, a finding opposite to the nature-activity interaction hypothesis (e.g., Maas et al. (2009)). People may not always react to the environmental cues. Individual preference and psychological dispositions seem to have a major role to play. As another example, it has been found that neighbourhood safety matters (Choi and Matz-Costa 2018; Kourtis et al., 2021). Yen et al. (2014) suggested that safety may affect mobility, which in turn determines physical health, social interactions and psychological wellbeing. However, while safety could have an impact on various successful ageing outcomes, our findings show that the effect of safety could be independent of social interactions, a finding consistent with Scott et al. (2018).

6. Conclusion

Identifying relevant environmental factors for the older people is a crucial task in the design of an urban or a suburban space in an ageing society (Milton et al., 2019). This study identified neighbourhood characteristics that contribute to successful ageing. Using a representative sample of 25,742 adults from 33 European countries, we found that health care services (HCS) affordability is a prominently relevant factor and can be more important than quality in predicting ageing outcomes in all age groups, a novel finding which has not been stressed in the literature. We also found that older people achieve better health and ageing outcomes in neighbourhoods which are safer and have better access to cultural and recreational facilities and green space. Geography certainly matters for successful ageing, but not all spatial-contextual factors are equally important.

This study has some limitations. First, the regression tree method is designed to identify patterns from data. They should not be interpreted as causal relationships. Second, self-esteem and other psychosocial factors are important to successful ageing (Bowling and Dieppe 2005). Although we have included them in our preliminary analysis, there are not that many relevant variables in the data set which may explain this aspect of ageing. In our initial analysis, neither personal factors (e.g., employment) nor activities (e.g., volunteering) play a role in predicting ageing outcomes. New data are needed to cover this knowledge gap. Third, while a machine-learning technique allows us to include a greater number of variables in an analysis, similar to a regression analysis, the method is prone to the missing data problem. How this missing data problem may influence our findings, however, is less known and is another topic for future research. Fourth, all wellbeing indicators are self-reported and subject to misreporting. Fifth, results reported in various Figures are based on group averages. Individual preference to certain environmental factors (e.g., nature) may play a role. As in the case of regression analysis, we simply rely on the random sample property of the sample: since the survey is nationally representative and the ageing outcomes are not selected based on certain outcomes and hence not systematically biased towards certain results, heterogeneity among individuals is likely to be averaged out and not seriously bias against our findings. Finally, limited variability in the variables may also affect our findings as the multivariable regression method has mainly been used in research using count measures as the outcome variables. However, the approach (i.e., minimisation of a distance measure) used to identify clusters is a commonly employed algorithm in cluster analysis, so it should be applicable to a continuous or an ordinal measure.

6.1. Relevance for policy

Despite some limitations in our empirical database, our study suggests on evidence-based grounds that resource deprivation can lead to poorer ageing outcomes. Consequently, the issue of fiscal resources allocation is critical when inequality among older adults has become an alarming public issue. As the affordability of health facilities, neighbourhood safety, and accessibility to cultural facilities and to green areas are major determinant of ageing outcomes, future policy should direct more resources to improve the affordability and quality of health care services, policing services, cultural facilities and urban greening in order to enhance the wellbeing of the ageing population.

Declarations of competing interest

None.

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